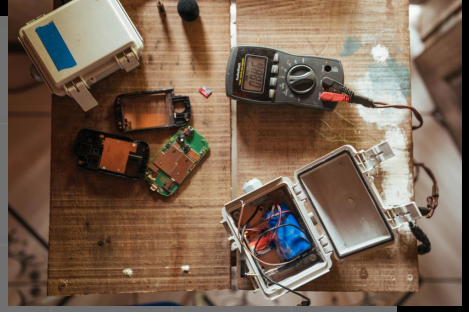
Leprosy of the land





THOUSANDS HECTARES OF LAND TURNED INTO LUNAR LANDSCAPES DUE TO ILLEGAL AMBER MINING. OUR MACHINE MODEL FOUND TRACES OF SUCH ACTIVITY ON SATELLITE IMAGES FOR THE AREA OF 70,000 SQUARE KM.

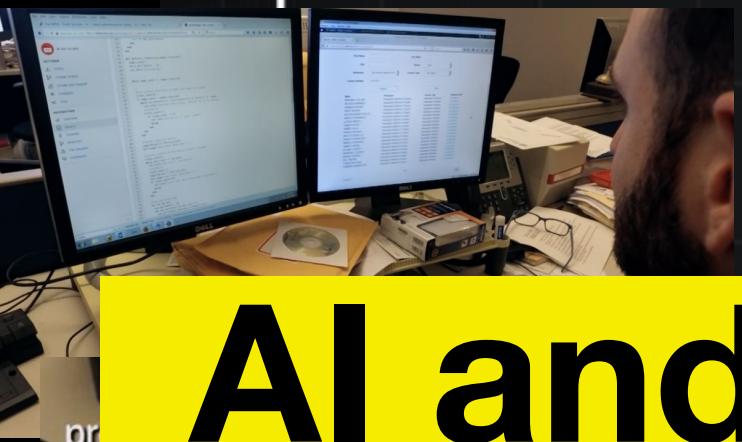
OPERAÇÃO

SERENATA DE AMOR

INTELIGÊNCIA ARTIFICIAL PARA CONTROLE SOCIAL DA ADMINISTRAÇÃO PÚBLICA

```
if(donation signal > 69) {
    if(state == 21) {
     return cluste
    } else {
     return cluster<19548>
    }
} else {
    if(age < 44) {
```

Manufacturers classified events "malfunction" even when patients died



"In the matter of restoring Internet freedom. I'd like to recommend the commission to undo The Obama/Wheeler power grab to control Internet access. Americans, as opposed to Washington bureaucrats,

AI and investigations

@PaulBradshaw

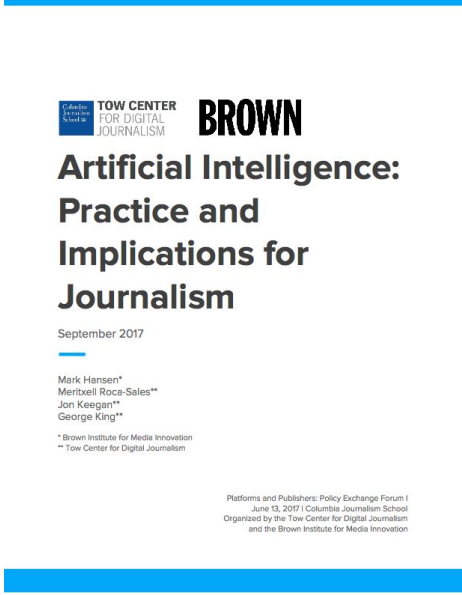
#dataharvest

Drag the sliders to predict well abandonment probability

bach that functioned very, very smoothly for

SciCAR 2019

3 applications of AI for newsrooms


TOW CENTER FOR DIGITAL JOURNALISM
BROWN
Artificial Intelligence: Practice and Implications for Journalism
September 2017
Mark Hansen*
Meritxell Roca-Sales**
Jon Keegan**
George King**
* Brown Institute for Media Innovation
** Tow Center for Digital Journalism
Platforms and Publishers: Policy Exchange Forum I
June 13, 2017 I Columbia Journalism School
Organized by the Tow Center for Digital Journalism
and the Brown Institute for Media Innovation

- Finding needles in haystacks
- Identifying trends (or departures from trends)
- Examining an application of AI or computation as the subject of the story itself

Hansen et al 2017

HOW ABOUT

1 1 ?

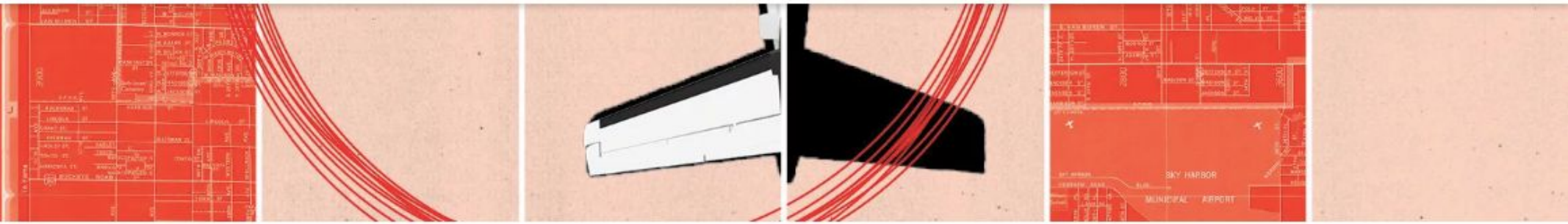
1. Establishing the scale of a problem*



At that point, our data journalism team wrote computer programs to "crawl" regulators' websites - a process known as scraping - and obtain board orders. This required building about 50 such programs tailored to agencies across the country. That collected more than 100,000 disciplinary documents. To assist us in identifying those involving sexual misconduct, we then created a computer program based on "machine learning" to analyze each case and, based on keywords, give each a probability rating that it was related to a case of physician sexual misconduct.

Atlanta Journal-Constitution

Training needed!



SCIENCE

We Trained A Computer To Search For Hidden Spy Planes. This Is What It Found.

From planes tracking drug traffickers to those testing new spying technology, US airspace is buzzing with surveillance aircraft operated for law enforcement and the military.



Peter Aldhous
BuzzFeed News Reporter

Last updated on August 8, 2017, at 10:47 a.m. ET

For the machine learning, we selected the random forest algorithm, popular among data scientists for classification tasks. (See this tutorial for background on running the random forest in R.)

As training data, drawn from `planes_features.csv`, we used 97 fixed-wing FBI and DHS planes from our previous story, given a `class` of `surveil`, and a random sample of 500 other planes, given a `class` of `other`.

Data identifying these planes is in the file `train.csv`.

BuzzFeed



Over the course of the Implant Files investigation, ICIJ and its partners filed more than 1,500 public records requests and collected more than 8 million device-related health records worldwide. These include recall notices, safety warnings, legal documents and corporate financial filings.

- Repeated classification of patient deaths as less serious incidents. Manufacturers and others required to report adverse events have classified more than 2,200 episodes in which patients died not as deaths but as injuries, malfunctions or other less severe events in the last five years.

The machine learning programs Talend Real-time BigData Platform, Microsoft SQL Server 2017 as well as the programming language R were used at different stages of the analysis. The entire process involved text mining, clustering, feature selection, association rules and classification algorithms to identify events not always described consistently in different parts of the data.

Machine learning algorithms were used, for example, to screen millions of records and identify reports in which the description of an adverse event indicated that a patient had died, but the death was misclassified as malfunctions or injuries.

NEWSLETTER

ICIJ / methodology

We tracked 25,688 abusive tweets sent to women MPs – half were directed at Diane Abbott

The abuse was measured over a six-month period.

By Azmina Dhrodia

		No. of Abusive Tweets	Percentage of All Abusive Tweets	M
Whole Period				
	1	8121	31,61%	**Diane Abb**
	2	1025	3,9%	Joanna Ch
	3	1023	3,9%	Emily Thor
	4	1002	3,9%	Jess Phillip
	5	875	3,4%	Anna Soub
6 Weeks Before				
	1	4512	45,14%	**Diane Abb**
	2	458	4,58%	Emily Thor
	3	368	3,68%	Joanna Ch
	4	347	3,47%	Amber Ru
	5	303	3,03%	Angela Ray

https://www.newstatesman.com/science-tech/2017/09/we-tracked-25688-abusive-tweets-sent-women-mps-half-were-directed-diane-abbott

Scale of abuse of politicians on Twitter revealed

9 November 2022



https://www.amnesty.org.uk/online-violence-women-mps

2. Proving a problem exists



Random Forest algorithm: SRF Data (GitHub repo): https://www.srf.ch/radio-srf-virus/

LAPD underreported serious assaults, skewing crime stats for 8 years



https://www.latimes.com/local/cityhall/la-me-crime-stats-20151015-story.html

3. Unmasking a system

How ProPublica's Message Machine Reverse Engineers Political Microtargeting

How the message machine works.

by Jeff Larson, Oct. 18, 2012, 8:33 a.m. EDT

Decision Trees are useful at finding hidden models in a particular data set, particularly because they produce a human-readable tree of partitions of the data. For example, here is a representation of the Decision Tree model generated for a single Obama for America message:

```
if(donation signal > 69) {
   if(state == 21) {
    return cluster<19546>
   } else {
    return cluster<19548>
   }
} else {
   if(age < 44) {
      if(donation signal > 64) {
        return cluster<19548>
      } else {
        return cluster<19546>
      }
   } else {
      return cluster<19545>
   }
}
```

https://www.propublica.org/nerds/how-propublicas-message-machine-reverse-engineers-political-microtargeting

4. Document histories: sentiment and other clues

Investigations

Whistleblowers say USAID's IG removed critical details from public reports

The Post obtained draft versions of 12 audits by the inspector general's office, covering projects from the Caribbean to Pakistan to the Republic of Georgia between 2011 and 2013. The drafts are confidential and rarely become public. The Post compared the drafts with the final reports published by the inspector general's office and interviewed former and current employees. E-mails and other internal records also were reviewed.

The Post tracked changes in the language that auditors used to describe USAID and its mission offices. The analysis found that more than 400 negative references were removed from the audits between the draft and final versions.

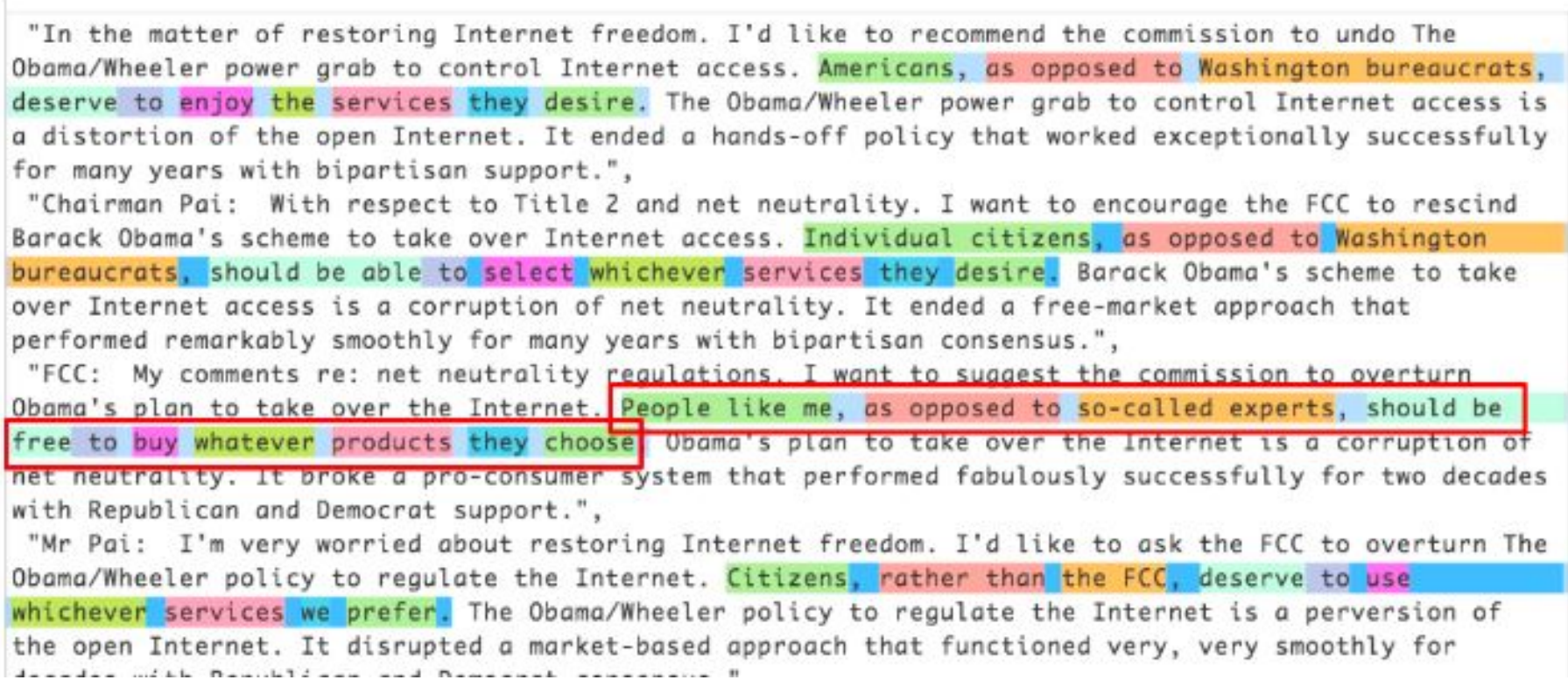
In one audit, the number of negative references fell from 113 to 61; in another, from 170 to 13.

https://www.washingtonpost.com/investigations/whistleblowers-say-usaids-ig-removed-critical-details-from-public-reports/2014/10/22/68fbc1a0-4031-11e4-b03f-de718edeb92f_story.html

5. Unmasking pseudo-human behaviour

More than a Million Pro-Repeal Net Neutrality Comments were Likely Faked

I used natural language processing techniques to analyze net neutrality comments submitted to the FCC from April-October 2017, and the results were disturbing.



"In the matter of restoring Internet freedom. I'd like to recommend the commission to undo The Obama/Wheeler power grab to control Internet access. Americans, as opposed to Washington bureaucrats, deserve to enjoy the services they desire. The Obama/Wheeler power grab to control Internet access is a distortion of the open Internet. It ended a hands-off policy that worked exceptionally successfully for many years with bipartisan support.",

"Chairman Pai: With respect to Title 2 and net neutrality. I want to encourage the FCC to rescind Barack Obama's scheme to take over Internet access. Individual citizens, as opposed to Washington bureaucrats, should be able to select whichever services they desire. Barack Obama's scheme to take over Internet access is a corruption of net neutrality. It ended a free-market approach that performed remarkably smoothly for many years with bipartisan consensus.",

"FCC: My comments re: net neutrality regulations. I want to suggest the commission to overturn Obama's plan to take over the Internet. People like me, as opposed to so-called experts, should be free to buy whatever products they choose. Obama's plan to take over the Internet is a corruption of net neutrality. It broke a pro-consumer system that performed fabulously successfully for two decades with Republican and Democrat support.",

"Mr Pai: I'm very worried about restoring Internet freedom. I'd like to ask the FCC to overturn The Obama/Wheeler policy to regulate the Internet. Citizens, rather than the FCC, deserve to use whichever services we prefer. The Obama/Wheeler policy to regulate the Internet is a perversion of the open Internet. It disrupted a market-based approach that functioned very, very smoothly for

Kao, 2017

6. Creating a basis for further enquiry

AP

Correction: Guns in School-Accidents story

Click to copy

RELATED TOPICS

AP Top News
Minnesota
North America
Gun politics
U.S. News
National Rifle Association

By RYAN J. FOLEY and LARRY FENN

Associated Press

They are the "good guys with guns" the National Rifle Association says are needed to protect students from shooters: a school police officer, a teacher who moonlights in law enforcement, a veteran sheriff.

Yet in a span of 48 hours in March, the three were responsible for gun safety lapses that put students in danger.

The data journalism team at AP, for example, used unsupervised machine learning to find hidden patterns in a data set of 140,000 human-entered incidents documented by the Gun Violence Archive (GVA). After discovering a host of errors in the initial data, AP used unsupervised machine learning to simplify the data and flag certain entries for further review without specific guidance.

AP was able to surface new story ideas that otherwise could have been missed. Journalists were able to find the "most typical" incidents related to gun violence, including whether the shooting was accidental or not, if it involved a child or even if a police officer was involved.

https://www.apnews.com/08659d568d7448a6b500f27d98a6c3a6
https://insights.ap.org/uploads/images/the-future-of-augmented-journalism_ap-report.pdf



What 1.2 million parliamentary speeches can teach us about *gender representation.*

With gender parity, constituents might hope that elected officials focus on mor inclusive issues. To better understand how a gender shift in the House of Commons might affect the topics debated, consider the speeches made by MP in the main chamber.

We compiled a collection of of 1.2 million speeches delivered by men and women who have served in the House of Commons since 1970, including monologues, debates, single-sentence replies, and acknowledgments.

We then used a probabilistic machine learning model[1] to help us infer the topics within a speech, looking for words that often appear together and clustering them into the same group[2] (eg. NHS & social care, energy policy, armed forces, education, etc.)[3]

https://pudding.cool/2018/07/women-in-parliament/



hosted by Open Knowledge Brazil using algorithms and AI to audit public spending and politicians expenses.[35] [36] The system uses a popular machine-learning toolkit for the open source python language to classify incoming expense records and identify outliers that it flags for human investigation. To date, the tool has identified over 8 000 suspicious payments, totalling BRL 3.6 million. During 2017, the project ran a hackathon event to carry out a "citizens' audit" of suspicious expense claims and generated over 200 reports to the official auditing body.[37] In total, the tool has processed more than 1.5 million expense claims.

https://stateofopendata.od4d.net/chapters/issues/artificial-intelligence.html

7. Jargon and obfuscation

School report

Holywell School

Red Lion Close, Cranfield, Bedford, MK43 0JA

Inspection dates	9–10 October 2014

Overall effectiveness	Previous inspection:	Not previously inspected	
	This inspection:	**Good**	**2**
Leadership and management		Good	2
Behaviour and safety of pupils		Good	2
Quality of teaching		Good	2
Achievement of pupils		Good	2

Summary of key findings for parents and pupils

This is a good school.

- Students in Year 8 achieve standards above the levels expected for their age, making good progress because they work hard and are well taught.
- Achievement in Key Stage 2 is good; students' progress has improved over time in reading, writing and mathematics.
- Disabled students and those who have special educational needs make good progress because the support provided is well-matched to their needs.
- Disadvantaged students achieve well because the extra funding to support them is used effectively.
- Students' behaviour and safety are good. They are keen to learn and to do well and there is a good working atmosphere in lessons. They say they feel safe. Students enjoy school, attend well and arrive punctually to their lessons.
- Support for students' spiritual, moral, social and cultural development is very effective.
- Teaching is good and improving because the school identifies and meets staff training needs well. Teachers use their good subject knowledge effectively to plan activities that interest and challenge students.
- The school offers a good range of subjects which promote students' academic and personal development well. Arrangements for tracking their progress are well-founded and used effectively to help them improve.
- The headteacher and senior leaders provide effective leadership within a caring school ethos. They track the progress students make regularly, monitor teaching carefully and have a clear and accurate picture of the school's strengths and development areas.
- Governors use their experience and expertise well to hold the school to account. They monitor the expenditure of school funds regularly and review information about students' performance carefully.

It is not yet an outstanding school because

- Not all teachers give clear guidance to students on how to improve their work or provide sufficient opportunities for students to act on it.
- Some teachers do not explain carefully enough what students are expected to learn in lessons.
- Not all staff fully implement the school's homework policy and some parents and students are not clear what is expected of them.
- There is not enough emphasis on improving students' grammar, spelling and presentation.

School report

Elm Tree Community Primary School

Elmers Wood Road, Tanhouse, Skelmersdale, Lancashire WN8 6SA

Inspection dates	26–27 September 2017

Overall effectiveness	**Outstanding**
Effectiveness of leadership and management	**Outstanding**
Quality of teaching, learning and assessment	**Outstanding**
Personal development, behaviour and welfare	**Outstanding**
Outcomes for pupils	**Outstanding**
Overall effectiveness at previous inspection	Good

Summary of key findings for parents and pupils

This is an outstanding school

- The headteacher's outstanding leadership ensures that pupils are at the heart of all that the school does.
- Believing in children is the watchword of leaders and staff. As a result, pupils with social, emotional and mental health needs thrive.
- Governors are passionate about the school and make a valuable contribution to school improvement.
- Pupils who attend Elm Tree have often struggled in their previous schools. Now all pupils are successful learners and the vast majority make outstanding progress from their starting points.
- Assessment is used well by teachers in lessons to match activities precisely to pupils' learning needs. This, together with strong, positive relationships and lots of praise and encouragement, ensures that pupils have an excellent attitude to learning.
- Pupils are offered a rich, broad and balanced curriculum. Physical education is particularly well taught. Pupils love this subject.
- The exceptional enrichment curriculum provides pupils with opportunities to participate in activities that they may not otherwise be able to experience. It is instrumental in building pupils' confidence, self-belief and social skills.
- The school is often innovative in its approach to working with pupils who have social, emotional and mental health needs. For example, assessing pupils' speech and language abilities and then building their skills up in this area has been instrumental in ensuring that pupils are ready to learn.
- Behaviour is outstanding. Pupils conduct themselves well around school and work hard to manage their own behaviour.
- Pupils' successful personal development and good welfare are central to the work of the school. Pupils benefit from a range of therapeutic support strategies such as play therapy, massage and counselling.
- Pupils feel very safe in school and parents agree.
- The school's outreach service has been instrumental in reducing the numbers of primary pupils excluded in the local authority.
- Sometimes the most able pupils are not challenged enough. At times they miss out on opportunities to think through mathematical problems and find answers for themselves.
- Occasionally, the end-of-year targets set for pupils are too aspirational and so it wrongly appears that some pupils do less well than they should.

8. Interesting associations, linking data

Locality Sensitive Hashing (LSH)

A visual description

https://artificialinformer.com/issue-one/dissecting-a-machine-learning-powered-investigation.html

A list of STRs registered with the city was the primary dataset. Secondarily, we got datasets of appraisal district properties and electric utility subscribers. We took the STR list and used a ML algorithm to look for matches in the other two datasets. This identified STR operators (individuals or couples) who owned multiple properties and had applied for exemptions.

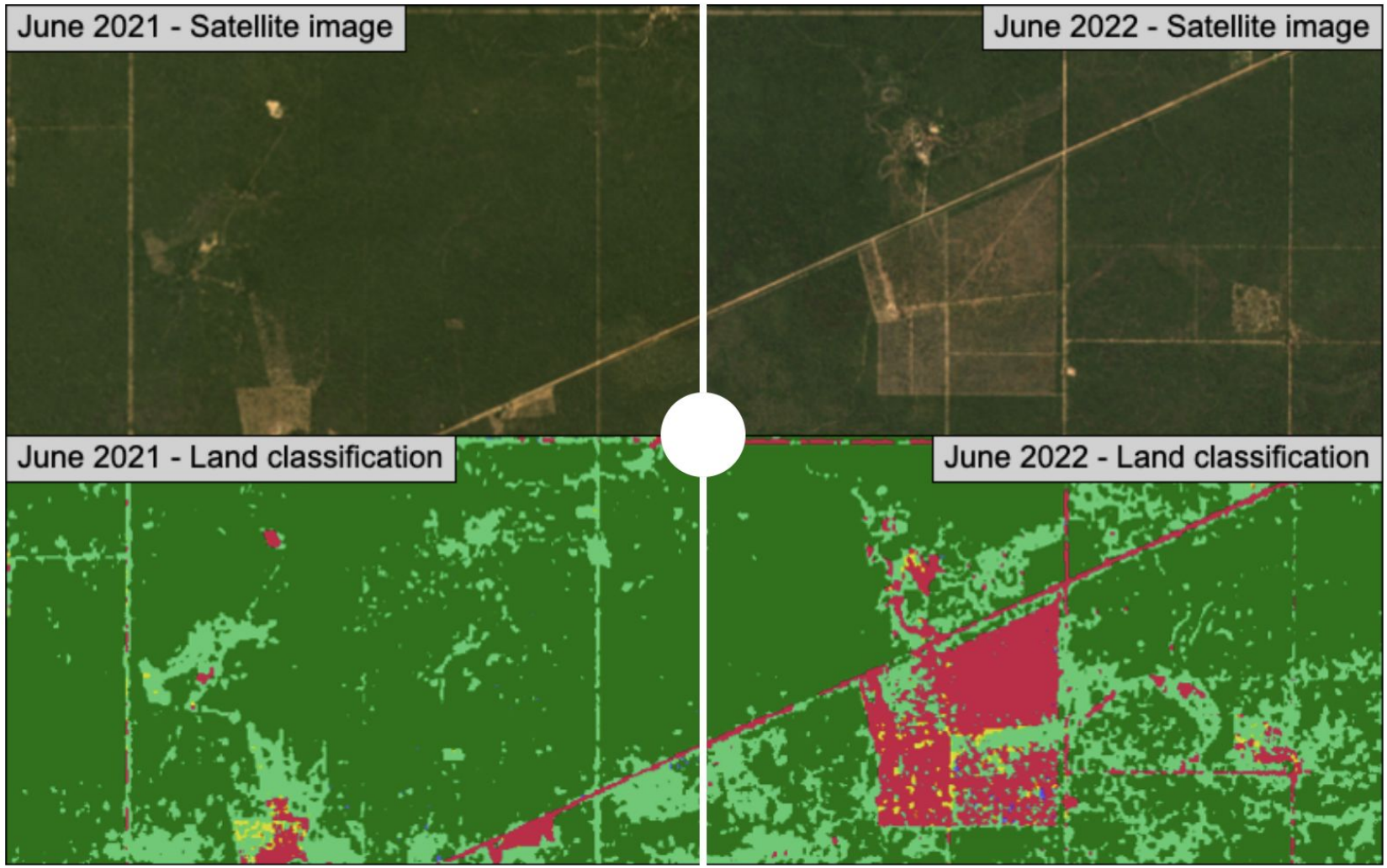
Next, we checked each joint STR/property record for exemptions and ranked them by suspiciousness. This ranking was accomplished using ML. From there, we filed a series of public information requests to verify our findings.

Joining two datasets based on one of their columns is a basic strategy common to many investigations. Relational databases were created to do exactly this. But in our case, we had low quality data and a lot of it: roughly a million records, combined. Names were spelled inconsistently and addresses were often abbreviated or were missing apartment numbers. Standard tools didn't cut it, so we turned to ML.

9. AI in the sky: satellite imagery, drone footage, sensors — and target acquisition

Use the slider to see how the AI identifies land use changes

By 2022, a patch changes from dark green to red in the algorithm's classification (bottom right), indicating possible deforestation. The original satellite image (top right) shows that the prediction is accurate.



Top: Copernicus Sentinel data [2022] processed by Sentinel Hub / Bottom: same images, processed through AI • Slider by Federico Acosta Rainis

https://facostarainis.github.io/ml-satellites/articles/chaco.html





Where There's Smoke

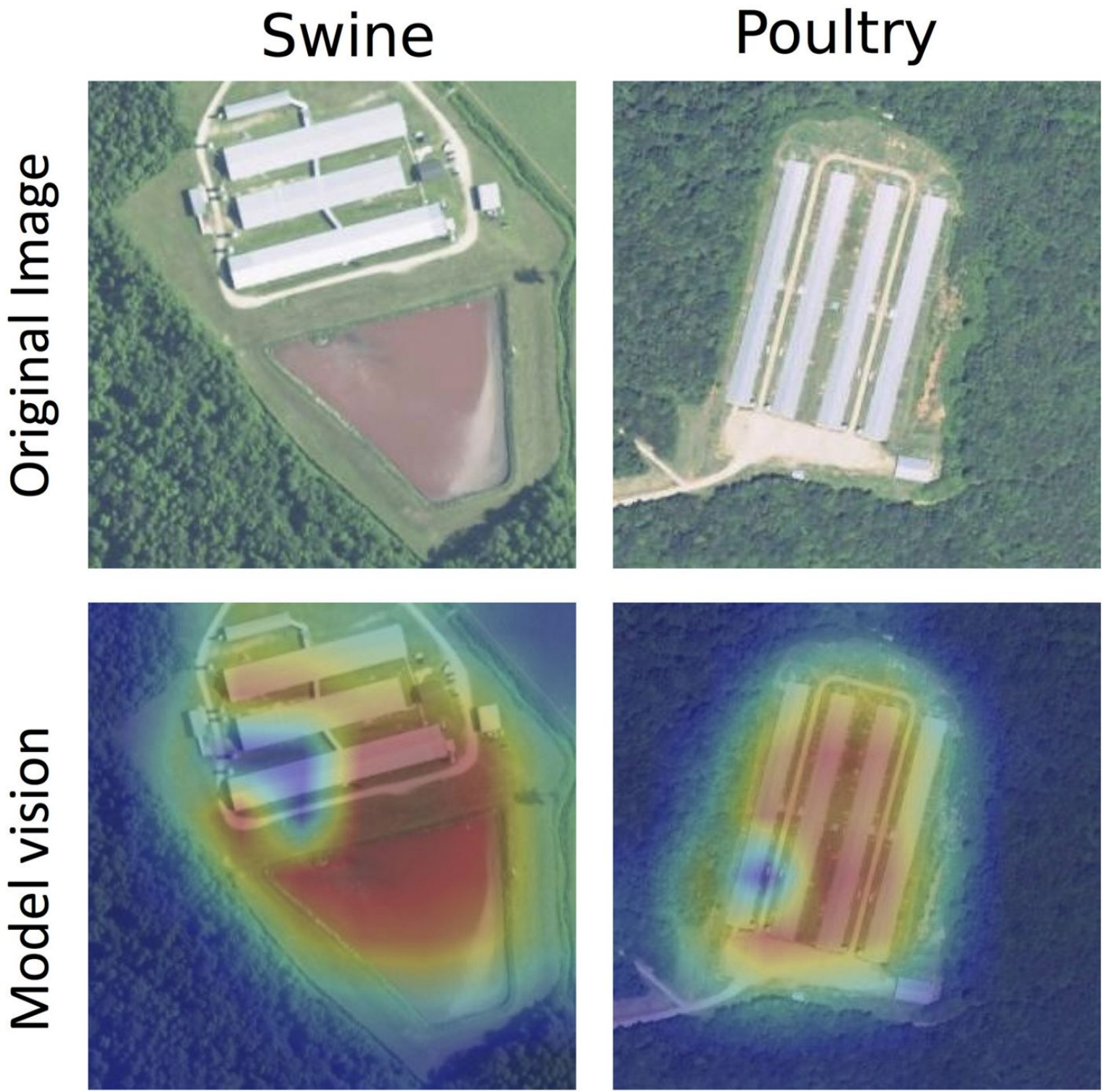
There are hundreds of satellites orbiting and observing the Earth. Aerial imagery can show geographic features, human settlements and forces like flood and fire. Comparing how a region looks at one moment in time compared to another can be critical for human rights investigations — but the influx of data is too vast for any individual to go through.

At GTC, Lyons shared how Human Rights Watch was able to use thermal data from environmental satellites to begin monitoring the outbreak of ethnic violence in Myanmar in 2017, just hours after the first reports of conflict. Combined with aerial images, the organization was able to detect a pattern of burned Rohingya villages across the region.

This digital evidence helped on-the-ground researchers corroborate the testimony of the Muslim minority community targeted by the authorities. By pinpointing the exact date and time that a village began burning, investigators could better quantify the scale of violence and begin to determine who the perpetrators were.

But it takes an expert eye — or a neural network — to tell the difference between smoke plumes and puffy white clouds.

https://blogs.nvidia.com/blog/2019/04/04/human-rights-watch-ai-gtc/



https://blogs.nvidia.com/blog/2019/04/04/human-rights-watch-ai-gtc/

click me



Leprosy of the land



THOUSANDS HECTARES OF LAND TURNED INTO LUNAR LANDSCAPES DUE TO ILLEGAL AMBER MINING. OUR MACHINE MODEL FOUND TRACES OF SUCH ACTIVITY ON SATELLITE IMAGES FOR THE AREA OF 70,000 SQUARE KM.

http://texty.org.ua/d/2018/amber_eng/

Google's artificial intelligence helps protect West Sumatra's rainforests



DYANING PANGESTIKA
THE JAKARTA POST

Jakarta / Sun, August 4, 2019 / 09:04 am



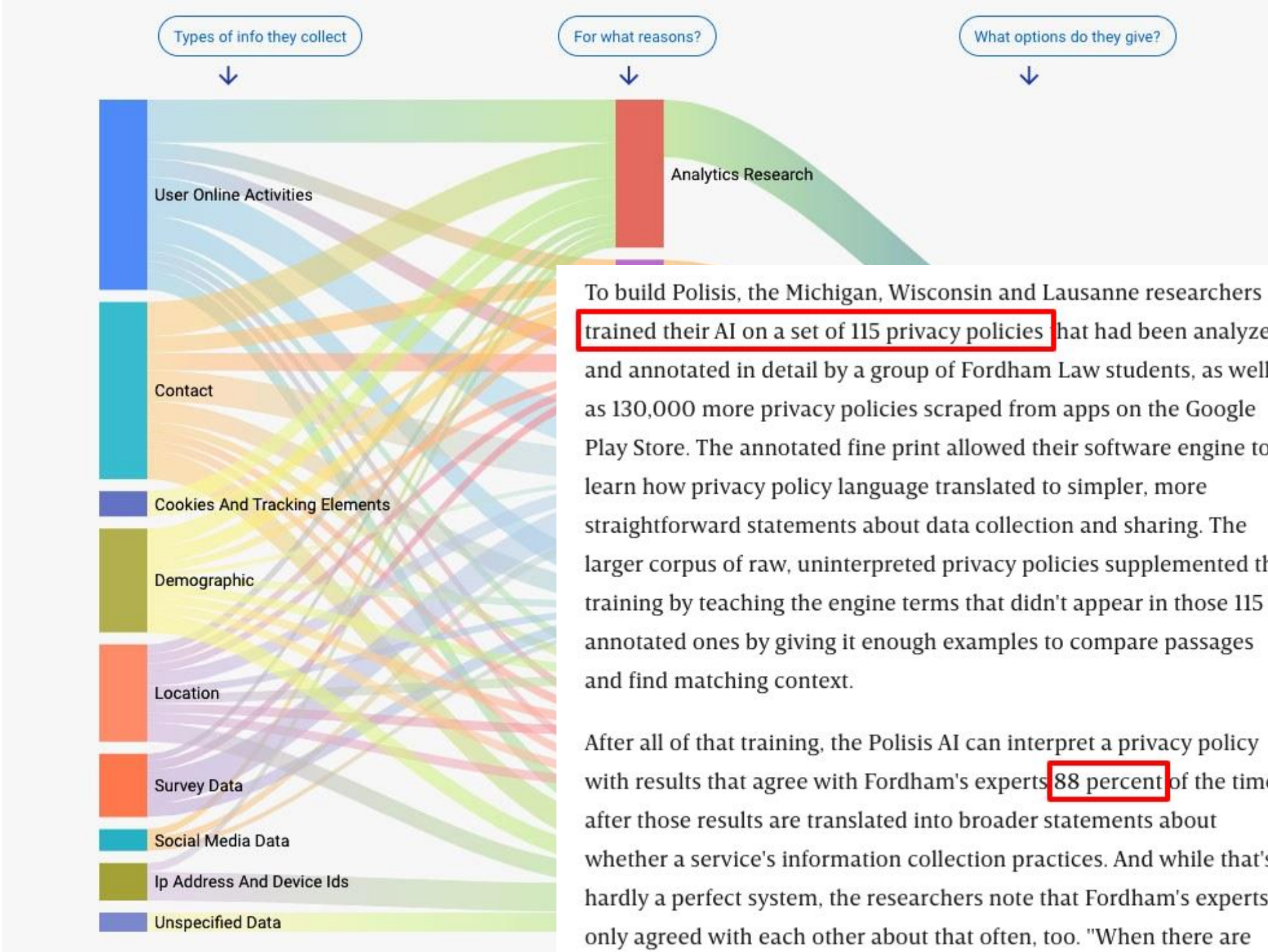
For instance, White said, "We should be able to detect animals that don't make sounds. Jaguars might not always be vocalizing, but the animals around them are, birds and things." The presence of a big cat then, might be easier to detect by listening for alarmed bird calls than for its near-silent movement through the forest.





https://www.thejakartapost.com/life/2019/08/02/googles-artificial-intelligence-helps-protect-west-sumatras-rainforests.html
https://techcrunch.com/2018/03/23/rainforest-connection-enlists-machine-learning-to-listen-for-loggers-and-jaguars-in-the-amazon/

10. Conversion: OCR, audio, video — and summaries



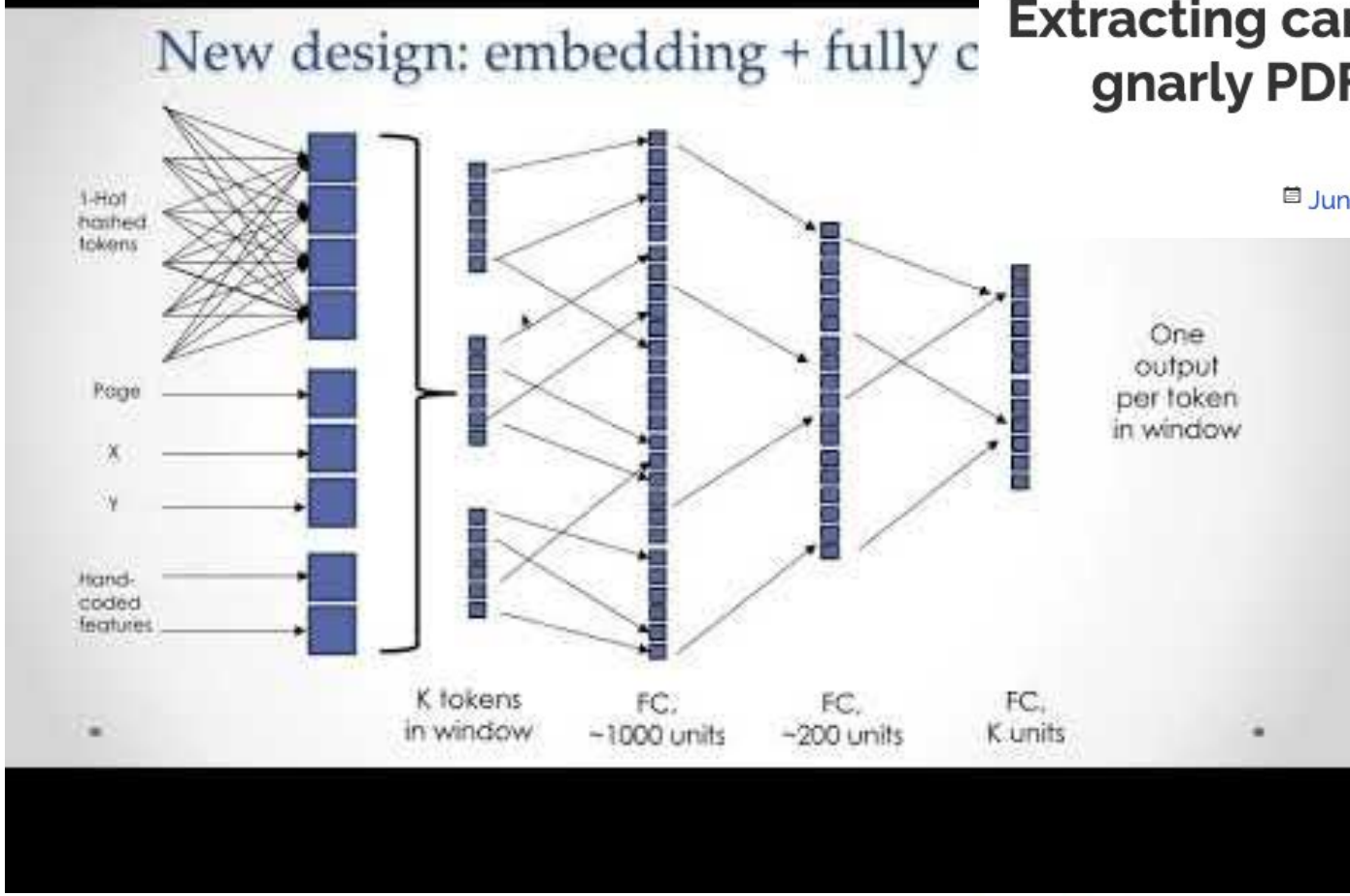
To build Polisis, the Michigan, Wisconsin and Lausanne researchers trained their AI on a set of 115 privacy policies that had been analyzed and annotated in detail by a group of Fordham Law students, as well as 130,000 more privacy policies scraped from apps on the Google Play Store. The annotated fine print allowed their software engine to learn how privacy policy language translated to simpler, more straightforward statements about data collection and sharing. The larger corpus of raw, uninterpreted privacy policies supplemented that training by teaching the engine terms that didn't appear in those 115 annotated ones by giving it enough examples to compare passages and find matching context.

After all of that training, the Polisis AI can interpret a privacy policy with results that agree with Fordham's experts 88 percent of the time, after those results are translated into broader statements about whether a service's information collection practices. And while that's hardly a perfect system, the researchers note that Fordham's experts only agreed with each other about that often, too. "When there are

https://www.wired.com/story/polisis-ai-reads-privacy-policies-so-you-dont-have-to/

Extracting campaign finance data from gnarly PDFs using deep learning

June 13, 2019 Jonathan Stray



In 2012, ProPublica ran the Free The Files project (you can read how it worked) and hundreds of volunteers hand-entered information for over 17,000 of these forms. That data drove a bunch of campaign finance coverage and is now available from their data store.

Can we replicate this data extraction using modern deep learning techniques? This project aimed to find out, and successfully extracted the easiest of the fields (total amount) at 90% accuracy using a relatively simple network.

http://jonathanstray.com/extracting-campaign-finance-data-from-gnarly-pdfs-using-deep-learning

11. Entry points: personalisation, A/B headlining, translation and chatbots

Climate change: Tree planting rise 'needs to happen quickly'

By Rob England
BBC News

30 July 2019

Share

Climate change



0,000 new trees but found it "painful" to get approval

12:10 30 Jul

257,900 government-funded trees planted in Cornwall in eight years

There have been 257,900 government-funded trees planted in Cornwall between 2010 and 2018, Forestry Commission data shows.

This works out at 459 trees per 1,000 people.

Between 2010 and 2018 the government funded about 15 million trees in England. Figures for trees planted with private funds or by local councils are not included and the data only covers new trees, not replacements for any that were cut down.

Overall tree planting rates across the UK would need to more than double to combat rising temperatures and climate change, according to official figures.

Last year 13,400 hectares of new trees were planted across the UK, most of them in Scotland, but the Committee on Climate Change says this should rise to at least 30,000 hectares a year.

Campaign group The Woodland Trust said more trees were needed to provide summer shade, reduce air pollution, improve water quality and manage flooding.

The government said it had made it "much easier" to apply for planting grants.

Find out about tree planting across the UK here.

This story has been generated using Forestry Commission data, BBC analysis and some automation.

https://github.com/BBC-Data-Unit/tree-planting



Where do you live, Alex?

https://www.akqa.com/work/lokai/walk-with-yeshi/

1. Establish scale
2. Reveal a system
3. Pseudo-human behaviour
4. Jargon
5. Associating data for linking
6. Identify leads
7. Multiple entry points for stories
8. Proving it exists
9. Document histories: sentiment
10. Satellite imagery
11. Conversion and summaries



Thank you.

@PaulBradshaw

#dataharvest